

LUREX: Propulsion and Launch Vehicle

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SUBSYSTEM OVERVIEW

Subsystem Objectives

- Launch Vehicle
- Qualify LiLO and MiLO as payloads on LV
 - Deliver both S.C. to TLI through NASA LSP
 - Define launch environment for all subsystems
 - Maintain schedule readiness (launch by Dec. 2030)

- Propulsion
- Insert into LLO and MLO lunar orbits respectively
 - Perform station-keeping and momentum dumping
 - Execute lunar crashdown at end of life

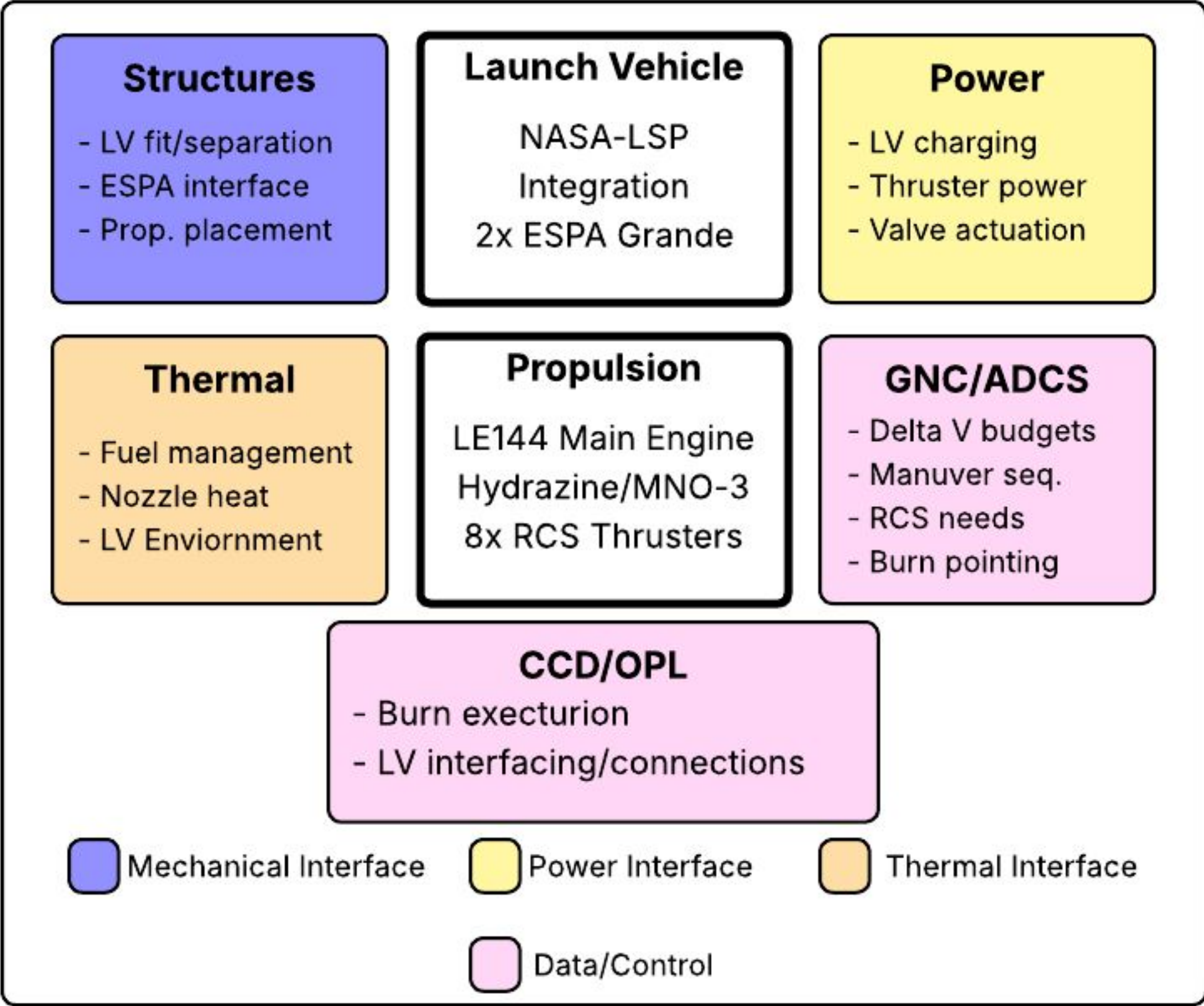
Subsystem Requirements

ID	Requirement	Value	Status
PLV-F-001	Apogee engine Isp	≥ 315 s	328 s
PLV-F-002	Apogee engine thrust	≥ 125 N	144 N
PLV-F-003	RCS thrust	≈ 1 N	1 N
PLV-P-004	LV characteristic energy	$C_3 = -1.8 \text{ km}^2/\text{s}^2$	Verified
PLV-C-006	Launch volume (per sat)	$42'' \times 46'' \times 56''$	Verified
PLV-C-007	Launch mass (per sat)	≤ 465 kg	300 kg
PLV-F-009	Launch deadline	Dec 26, 2030	Verified

Subsystem Drivers

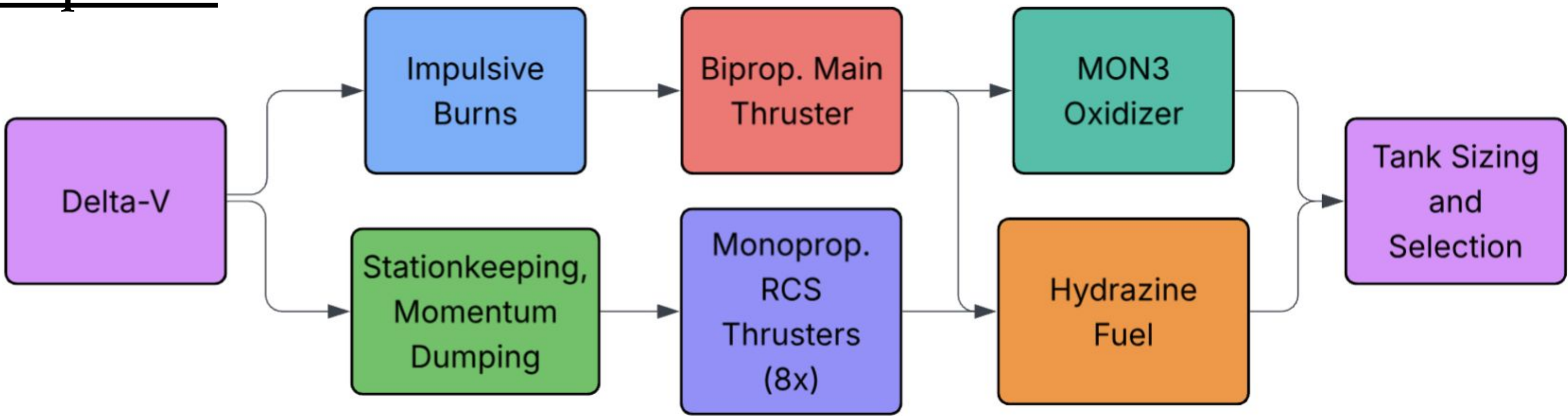
- ΔV Budget: Primary, drives all propellant and thruster sizing
- Simplicity: shared hydrazine tank for RCS and main thrusters - minimizes failure points
- Mass: Wet mass governs Tsiolkovsky’s rocket equation
- Volume: ESPA Grande constraints, tank and thruster packaging

Interactions



FORMAL ANALYSIS

Propulsion



Velocity Control Thruster Analysis

Thruster	Manufacturer	I_{sp} (s)	Thrust (N)	Fuel Type	Mass (kg)	Length	Heritage
S400-15	Ariane Group	321	425	MMH/MON-1	4.3	Unknown	Yes
TR501	Northrop Grumman	315	51	Hydrazine/ N_2O_4	2.4	12.25"	Yes
R-4D-11	Aerojet Rocketdyne	311	490	MMH/NTO	3.8	21.8"	Yes
R-4D-15	Aerojet Rocketdyne	326	445	Hydrazine/NTO	5.2	24.75"	Yes
BT-4	IHI Aerospace	329	450	Hydrazine/NTO	4.0	26"	Yes
LE144	Agile Space Industries	328	144	Hydrazine/MON3	1.8	13.90"	No

Propellant Load Analysis

Assumptions & Derived Quantities					
	Isp RCS (s)	Isp VCE (s)	g_0 (m/s ²)	m_dry (kg)	
	215.00	328.00	9.81	160.00	
	MT-9 datasheet	LE144 datasheet	Standard	Current SMM (dry only)	
MiLO Satellite			LiLO Satellite		
Maneuver	ΔV (m/s)	Propellant (kg)	Maneuver	ΔV (m/s)	Propellant (kg)
LOI Maneuver	590.00	41.84	LOI Maneuver	654.80	41.75
Approach Burn	110.60	7.02	LOI Maneuver	33.99	1.94
Orbital Insertian	121.90	7.47	Circularize	134.35	7.49
Station Keeping (total)	188.90	16.57	Inclination Change	250.76	13.16
n/a	0.00	0.00	n/a	0.00	0.00
Collision Avoidance	15.75	1.63	Collision Avoidance	15.75	1.21
Crash burn	298.10	15.50	Crash burn	31.26	1.56
TLI Corrections	15.75	1.19	TLI Corrections	15.75	1.19
RCS M.D.		0.05	RCS M.D.		0.05
Subtotal	1341.00	91.28	Propellant Load	1136.66	68.35
Derived Wet Mass	MiLO m_0 (kg)	251.275	Derived Wet Mass	LiLO m_0 (kg)	228.351

$m_0 = m_{dry} \cdot \exp(\sum \Delta V_{vce} / (I_{sp_vce} \cdot g_0)) \cdot \exp(\sum \Delta V_{rcs} / (I_{sp_rcs} \cdot g_0))$ | Propellant depleted sequentially burn-by-burn from m_0 | Blue = inputs | Green = derived outputs | Col G = internal m_0 (do not edit)

Launch Vehicle

Launch Option	Payload to TLI (kg)	Cost (M USD)	Reliability	Critical Details
NASA LSP Option 3 [80]	465	Mission allocation	Very High	NASA manages procurement, integration, and certification. Streamlines export control compliance.
Falcon 9 Rideshare* [103]	Limited (secondary payload)	~15.7 **	~98%	TLI rideshare available as secondary on lunar missions (e.g. CLPS). Mass and schedule set by primary payload.
Falcon 9 Dedicated* [104]	~1925-3520	~67	~98%	Strong performance margin for direct TLI. May be oversized for smaller spacecraft.
Vulcan Centaur Rideshare (VC0S) [105]	Limited (secondary payload)	~24.4 **	Moderate-High	Proportional cost estimate based on 465 kg of 2,100 kg VC0S TLI capacity at \$110M. Schedule and mass allocation set by primary payload.
Vulcan Centaur Dedicated (VC0S) [105]	~2100	~110	Moderate-High	High-performance cislunar vehicle. Limited heritage as a new system introduces schedule risk.
Rocket Lab Electron / Neutron [106]	Electron: ~25-40 Neutron: ~1500+	Electron: ~7.5 Neutron: TBD	Moderate	Electron undersized for mission. Neutron promising but remains in development with no flight heritage.

Table 74: LV Trade Study Summary

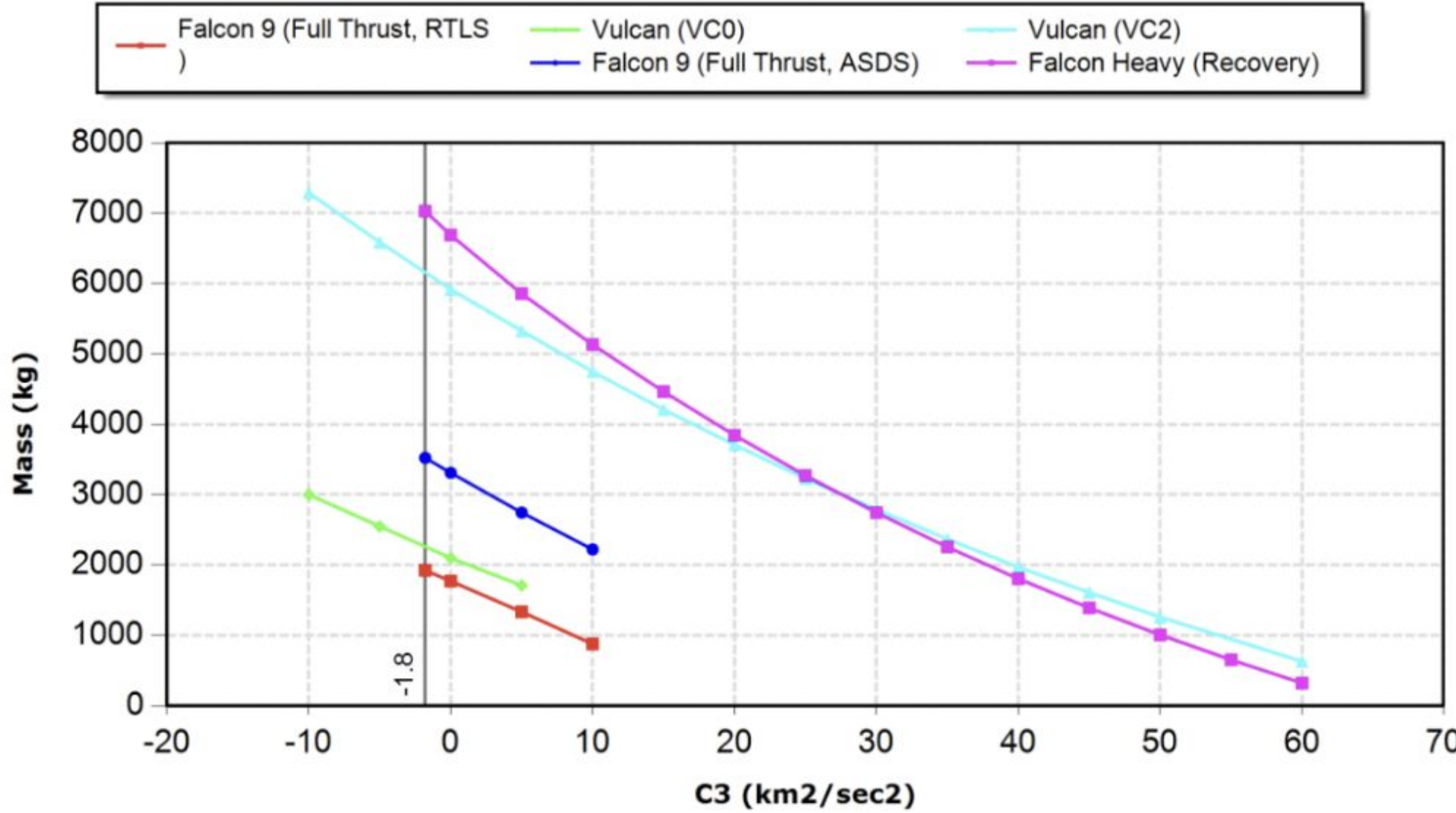
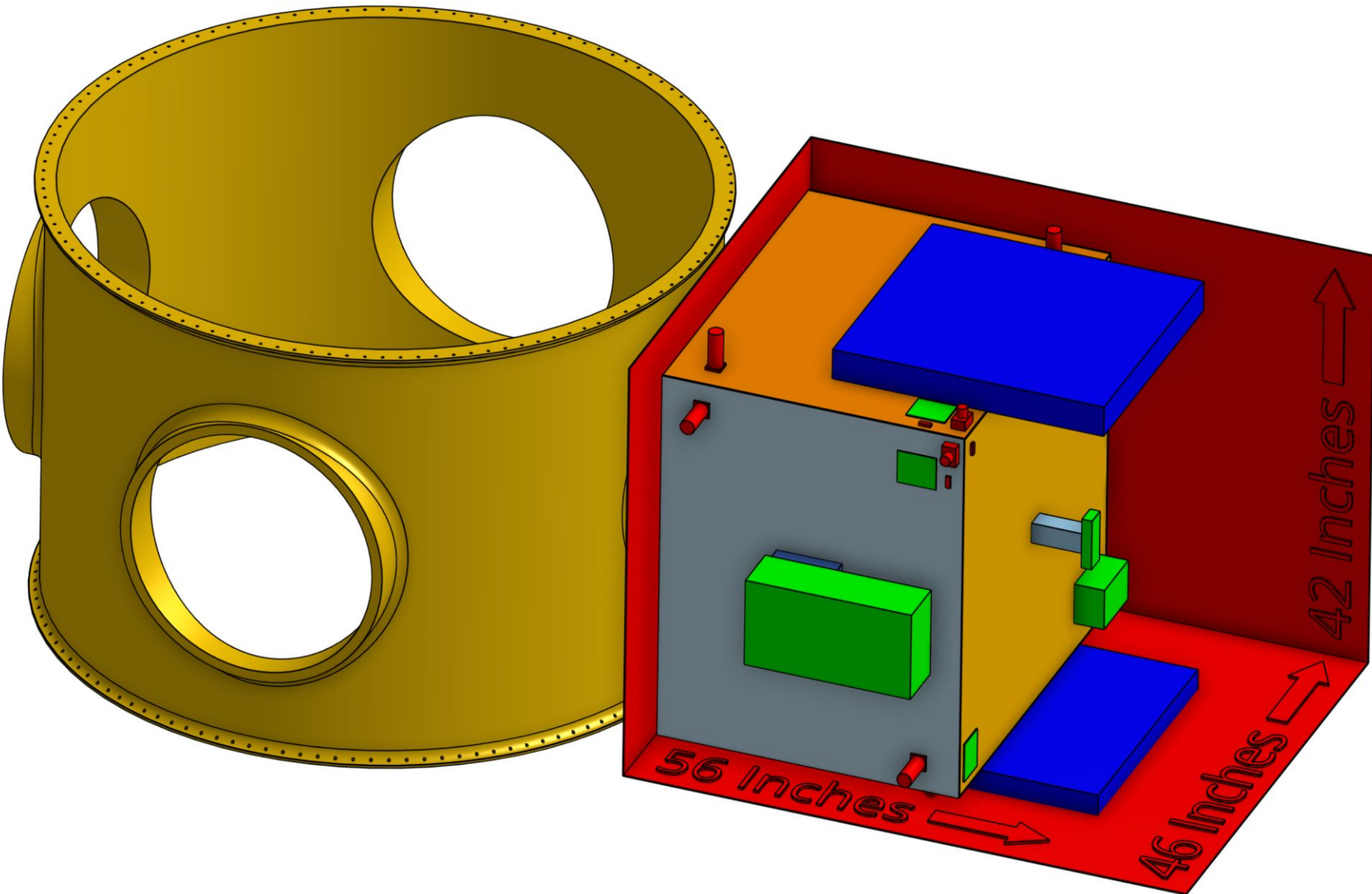


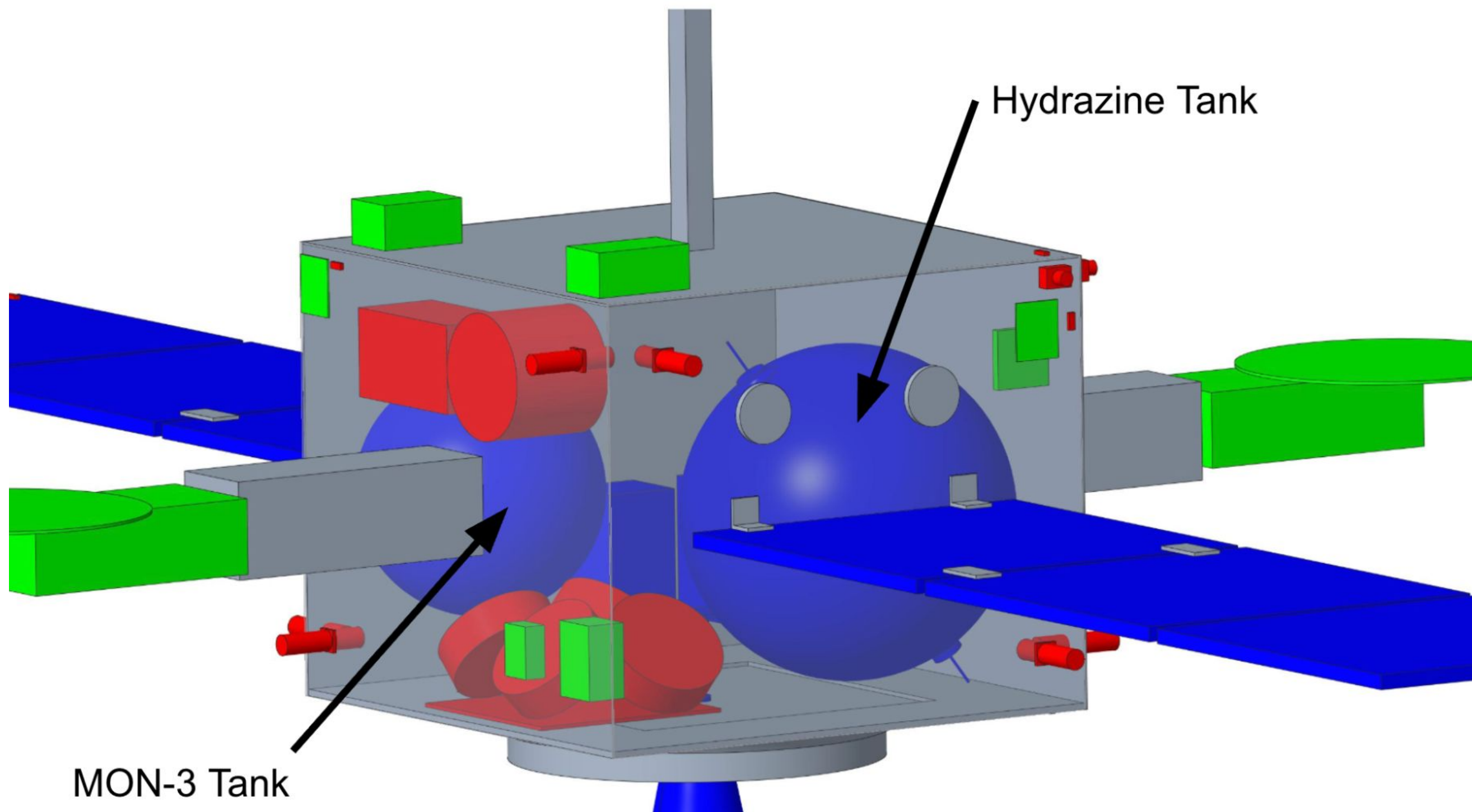
Figure 1: Launch Mass Performance per C3 by Launch Provider

FINAL DESIGN CONFIGURATION

Launch Vehicle



Propulsion System



VERIFICATION & VALIDATION

Table 77: Burn Time Impulsive Approximation Validation

Burn	ΔV (m/s)	m_0 (kg)	Burn Time (s)	Orbit Period (s)	Ratio
LiLO LOI	654.80	251.63	1034.16	17,651	5.85%
LiLO inclination change	250.76	200.36	335.66	7,065	4.75%
MiLO LOI	590	251.63	941.98	31,470	2.99%
MiLO EOL	298	174.90	345.80	86,217	0.54%